BASE CHANGE OF CLOSED SUBSETS PROPER OVER THE BASE: IN A CARTESIAN SQUARE WITH F LOCALLY OF FINITE TYPE, THE INVERSE IMAGE OF A CLOSED SUBSET Z OF X PROPER OVER S IS A CLOSED SUBSET OF X' PROPER OVER S'.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7503-7515 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 7517-7526, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33h (tail batch, final screening of ranks 86-118).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01T4: sources/stacks-coherent/pinned-original-source.tex, source lines 7526–7526. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W4: sources/stacks-coherent/pinned-original-source.tex, source lines 7526–7526. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JV: sources/stacks-coherent/pinned-original-source.tex, source lines 7526–7526. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

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with f locally of finite type. If Z is a closed subset of X proper over S, then $(g')^{-1}(Z)$ is a closed subset of X' proper over S'.

Proof. Observe that the statement makes sense as f' is locally of finite type by Morphisms, Lemma 01T4. Endow Z with the reduced induced closed subscheme structure. Denote $Z' = (g')^{-1}(Z)$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z' = X' \times_X Z = (S' \times_S X) \times_X Z = S' \times_S Z$ is proper over S' as a base change of Z over S (Morphisms, Lemma 01W4). □